

Matteo Rossato

Software Engineer · Backend & Distributed Systems

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Italy - Open to Relocation & Remote

SUMMARY

Software Engineer with 3+ years building production-grade backend systems and data infrastructure at a Fortune 500 global manufacturer (AGCO, €12B+ revenue). Ship end-to-end pipelines over enterprise-scale data, backend automation platforms, and ETL integrated with ERP and cloud services. Strong foundations in algorithms, distributed systems, and system design. Pursuing an MSc in Computer Science at Politecnico di Milano. Seeking a software engineering role building reliable, scalable systems at global scale.

EXPERIENCE

Software Engineer - Data & Backend Systems
AGCO Corporation

2022 - Present
Italy · Fortune 500 manufacturer (€12B+ revenue)

- Architected and shipped an end-to-end batch data pipeline (Python, SQL) processing large-scale enterprise datasets across multiple ERP systems - **cut manual processing time by 25%**.
- Engineered a pipeline validation and monitoring framework with automated error-handling and alerting - **reduced missing-data incidents by 55%** across production workflows.
- Improved data processing throughput by **10%** through query optimization, indexing strategies, and architectural refactoring of legacy backend services.
- Built a multi-source backend platform unifying SQL, ERP, and third-party APIs into a single normalized layer with transformation and validation stages - **reduced data errors by 15%** and eliminated redundant manual workflows.
- Led technical decision-making for data architecture on **€25M+ engineering programs** - defined the engineering roadmap, evaluated build/buy tradeoffs, and aligned system design with business requirements.
- Designed a KPI tracking and reporting system used by cross-functional stakeholders across **four European business units**.

PROJECTS

Distributed Rate Limiter *Python · Redis · Lua · FastAPI · Docker*

- Built a distributed rate-limiting service supporting token-bucket and sliding-window algorithms, using atomic Redis operations (Lua scripts) for correctness under concurrent load.
- Exposed as configurable middleware / HTTP API with per-client limits; load-tested for high-throughput concurrent traffic.

MiniRAG - Retrieval-Augmented Generation Engine *Python · Embeddings · Vector Search · FastAPI*

- Implemented a compact RAG system: document chunking, embedding generation, vector similarity search, and LLM answer synthesis with source citations.
- Modular retriever/generator design with a pluggable vector store (FAISS / pgvector) and a retrieval-quality evaluation harness.

Automated Content Generation Pipeline *Python · Anthropic Claude API · WordPress REST API*

- Designed a modular AI pipeline producing **700+ SEO-optimized articles/month** end-to-end - topic selection, drafting, and CMS publishing.
- Fault-tolerant with retry logic, structured logging, and configurable stage execution; near-zero manual intervention in production.

EDUCATION

MSc Computer Science - Politecnico di Milano
Focus: systems, algorithms, applied machine learning

2026 - 2028 (in progress)

MSc Bioengineering - University of Padua

2018 - 2021

BSc Computer Engineering - University of Padua

2015 - 2018

SKILLS

Languages: Python (Advanced), SQL (Advanced), JavaScript, C++
Backend & Systems: REST APIs, ETL & Data Pipelines, Caching (Redis), Concurrency, System Design, Distributed Systems
Cloud & Tools: AWS, Docker, Git, CI/CD, PostgreSQL, NoSQL
Concepts: Algorithms, Performance Optimization, Scalability, Data Modeling

CERTIFICATIONS & LANGUAGES

Certifications: Google Data Analytics Certificate · English - C1 IELTS Academic
Languages: Italian (native), English (C1 - professional working proficiency)